SQL Syntax Cheat Sheet

Reference for Midterm Exam - SQLite

1. SELECT + WHERE + ORDER BY + LIMIT

```
SELECT col1, col2
FROM tbl1
WHERE condition
ORDER BY col1 DESC
LIMIT 10;
```

2. JOIN

```
SELECT tbl1.col1, tbl2.col1
FROM tbl1
JOIN tbl2 ON tbl1.id = tbl2.id;
```

3. GROUP BY + HAVING + Aggregate

```
SELECT col1, COUNT(*), AVG(col2)
FROM tbl1
GROUP BY col1
HAVING COUNT(*) > 5;
```

4. Subquery in WHERE

```
SELECT col1 FROM tbl1
WHERE col2 IN (
    SELECT col2 FROM tbl2
    WHERE condition
);
```

5. Subquery in HAVING

```
SELECT col1, COUNT(*)
FROM tbl1
GROUP BY col1
HAVING COUNT(*) > (
    SELECT AVG(col2) FROM tbl2
);
```

6. Subquery in FROM

```
SELECT col1, total
FROM (
    SELECT col1, COUNT(*) as total
    FROM tbl1
    GROUP BY col1
) as sub
WHERE total > 5;
```

7. Subquery in SELECT

```
SELECT col1,
    (SELECT COUNT(*) FROM tbl2
     WHERE tbl2.id = tbl1.id) as total
FROM tbl1;
```

8. Correlated Subquery

```
SELECT col1, col2
FROM tbl1 t1
WHERE col2 > (
    SELECT AVG(col2) FROM tbl1 t2
    WHERE t1.col1 = t2.col1
);
```

9. Window Function

```
SELECT col1,
    FUNCTION() OVER (
        PARTITION BY col1
        ORDER BY col2
        ROWS BETWEEN VALUE1 AND VALUE2
    ) as result
FROM tbl1;
```

10. CASE WHEN

```
SELECT col1,
    CASE
        WHEN condition THEN 'High'
        WHEN condition THEN 'Medium'
        ELSE 'Low'
    END as category
FROM tbl1;
```

11. DateTime (SQLite)

```
strftime('%Y', col1)  -- year
strftime('%m', col1)  -- month
strftime('%d', col1)  -- day
```

12. CTE (Common Table Expression)

```
WITH cte_name AS (
    SELECT col1, COUNT(*) as total
    FROM tbl1
    GROUP BY col1
)
SELECT * FROM cte_name WHERE total > 5;
```

13. ROWS BETWEEN - Possible Values

Value	Meaning
UNBOUNDED PRECEDING	Start of the partition (first row)
CURRENT ROW	The current row being processed
UNBOUNDED FOLLOWING	End of the partition (last row)
N PRECEDING	N rows before current row (e.g. 2 PRECEDING)
N FOLLOWING	N rows after current row (e.g. 1 FOLLOWING)